

Per-jet data/MC corrections for the $H \rightarrow \mu\mu$ analysis:
extraction of the hard- and pileup-jet weights and of the Drell–Yan
($p_T^{\ell\ell}, N_{\text{jet}}$) corrections from a joint binned maximum-likelihood fit,
with their covariance

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Abstract

The simulation used in the $H \rightarrow \mu\mu$ analysis mismodels the jet activity of the Drell–Yan background: forward jets are over-predicted by 20–40%, the number of jets falls too slowly and the $p_T^{\ell\ell}$ spectrum differs from data. We derive a set of universal per-jet weights, one for jets matched to a generator-level jet (“hard”) and one for unmatched (“pileup”) jets, on a grid of jet η and p_T , together with process-level corrections to the Drell–Yan ($N_{\text{jet}}, p_T^{\ell\ell}$) and W+jets N_{jet} distributions, from a single binned maximum-likelihood fit to 2024 data (109.99 fb^{-1}) in two control regions: $Z \rightarrow ee$ events split into categories of the jet–boson topology, and a single-muon $W \rightarrow \mu\nu$ selection. The likelihood is a Poisson likelihood of raw data counts with the Barlow–Beeston-lite treatment of the simulation statistics; the full covariance of all 772 parameters is obtained from the inverse Hessian and validated with a sandwich estimator that accounts for the events shared between the fitted histograms. The corrections are validated in $Z \rightarrow \mu\mu$ and in a $t\bar{t}$ -dominated $e\mu$ region, and the residual non-closure in these regions is turned into additional nuisance parameters. The method replaces an iterative weighted least-squares procedure, reproduces its closure where the data constrain the weights, and provides the first consistent propagation of the uncertainties of the maps and of the ($N_{\text{jet}}, p_T^{\ell\ell}$) corrections.

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1 Introduction

The $H \rightarrow \mu\mu$ analysis categorises events by their jet activity (gluon-fusion categories by N_{jet} and $p_{\text{T}}^{\ell\ell}$, the VBF category by the dijet kinematics), and the dominant background, Drell–Yan (DY) $Z/\gamma^* \rightarrow \mu\mu$, is taken from simulation shaped by data in the Z mass window. The 2024 simulation (amc@NLO FxFx DY, NanoAODv15 with the 2024 jet energy corrections and the Run 3 pileup profile) does not describe the jet activity of the data: the per-jet data/MC ratio drops to 0.6–0.8 for $|\eta| > 2.5$ at low p_{T} , there is a localised excess (“horn”) at $|\eta| \approx 3$, and the N_{jet} distribution has a slope of $\sim 5\%$ per jet. Earlier work established that the effect is driven by jets that are not matched to a generator-level jet (pileup, “PU” jets) and that a per-jet correction as a function of (η, p_{T}) with separate factors for matched (“hard”) and unmatched jets transports between processes with different hard/PU compositions, while a single all-jet weight does not.

This note documents the extraction of those weights with a proper likelihood. The previous procedure (“wzll”) solved, in every (η, p_{T}) cell, a two-parameter weighted least-squares problem over the control-region rows with hand-set relative weights of the regions, Gaussian pseudo-measurements as priors, data-only variances, and alternated it with an exact step for the DY ($N_{\text{jet}}, p_{\text{T}}^{\ell\ell}$) correction; its per-cell uncertainties neglected the correction factors and the correlations between cells. The present method keeps the same ingredients (regions, categories, observables, pileup reweighting, gauge convention) and replaces the statistical treatment: one Poisson likelihood over the jet and event tables of the two control regions, all corrections as free parameters, simulation statistics in the likelihood, and the full covariance propagated to the validation histograms.

2 Data, simulation and object selection

Data. The full 2024 dataset (eras C–I, 109.99fb^{-1} after the certified-lumi selection, brilcalc) in the Muon0/1 and EGamma0/1 primary datasets. Trigger: HLT_IsoMu24 for the muon datasets; HLT_Ele30_WPTight_Gsf and not HLT_IsoMu24 for the electron datasets, so that no data event enters two channels. One lepton is required to match a trigger object (muon $p_{\text{T}} > 26\text{ GeV}$ matched to a muon trigger object; electron $p_{\text{T}} > 33\text{ GeV}$ with the WPTight filter bit). In the single-muon W selection the data are prescaled by a factor 20 at the dump stage (every 20th event is kept); this prescale is treated as an exposure factor in the likelihood, never as an event weight.

Simulation. NanoAODv15 Run 3 2024 samples: DY $\rightarrow ee/\mu\mu/\tau\tau$ ($M_{\ell\ell} > 50\text{ GeV}$, amc@NLO FxFx), EWK $Z + 2$ jets (MadGraph+Herwig), $W \rightarrow e\nu/\mu\nu/\tau\nu$ (amc@NLO FxFx), $t\bar{t}$ (dileptonic, semileptonic, hadronic), single top (t - and s -channel, tW), and diboson WW/WZ/ZZ (POWHEG). Events are weighted by the generator weight, the pileup weight, the lepton identification, isolation and trigger scale factors (muon tight ID and tight PF isolation with the corresponding IsoMu24 scale factors in the single-muon selection; electron MVA-iso WP90 ID and Ele30 HLT scale factors), and normalised to the cross section and luminosity. Four “families” are used in the fit: DY, W, Top ($t\bar{t}$ and single top) and “other” (diboson and EWK Z).

Leptons. Muons: beam-spot-constrained p_{T} where $\chi_{\text{BS}}^2 < 30$, $|\eta| < 2.4$, medium ID, PF isolation $\text{ID} \geq 2$ (dilepton selections) or tight ID and tight PF isolation (single-muon selection), $p_{\text{T}} > 10\text{ GeV}$ (leading muon $p_{\text{T}} > 26\text{ GeV}$). Electrons: $|\eta| < 2.5$, MVA-iso WP90, EGM scale (data) and smearing (simulation) corrections applied, corrected $p_{\text{T}} > 20\text{ GeV}$. Dilepton events require exactly two good leptons of opposite charge; the single-muon selection requires exactly one good muon and no additional loose lepton.

Jets. AK4 PUPPI jets with the 2024 jet energy corrections (V5 L1L2L3 and residuals) and the 2024 resolution smearing in simulation, $p_{\text{T}} > 25\text{ GeV}$, $|\eta| < 4.7$, tight jet ID, no overlap with a good lepton ($\Delta R < 0.4$), and the *horn veto*: jets with $2.5 \leq |\eta| < 3.0$ and $p_{\text{T}} < 50\text{ GeV}$ are removed from the analysis jet collection (they are stored separately and are not used here). A jet in simulation is

hard when it is matched to a generator-level jet ($\text{Jet_genJetIdx} \geq 0$, $\Delta R < 0.4$) and *PU* otherwise; the flag is not defined for data. *b*-tagged jets use the medium working point of the ParticleNet tagger. The analysis-level MET is PUPPI MET with the jet energy corrections and the lepton momentum corrections propagated (“corrected MET”).

3 Control regions, categories and observables

Table 1 lists the regions. Two of them enter the fit; the other two are used for validation and for the non-closure nuisances. In every dilepton region the dijet “VBF” subset (≥ 2 jets, $m_{jj} > 400$ GeV, $|\Delta\eta_{jj}| > 2.5$ for the leading pair) is also monitored.

Table 1: Regions. “*b*-veto” = no medium *b*-tagged jet. In the fit regions the data prescale is 1 for *ee-Z* and 20 for μ -*W*.

region	channel	role	selection
<i>ee-Z</i>	<i>ee</i>	fit	$70 < m_{ee} < 110$ GeV, <i>b</i> -veto; jets split into five jet–boson categories (Table 2)
μ - <i>W</i>	single μ	fit	tight μ , $m_T(\mu, \text{MET}_{\text{corr}}) > 50$ GeV, $\text{MET}_{\text{corr}} > 30$ GeV, no extra loose lepton, <i>b</i> -veto
$\mu\mu$ - <i>Z</i>	$\mu\mu$	validation	$70 < m_{\mu\mu} < 110$ GeV, <i>b</i> -veto (the analysis channel)
<i>eμ</i> -Top	<i>eμ</i>	validation	$m_{e\mu} > 50$ GeV, ≥ 1 medium <i>b</i> jet; observables built from the non- <i>b</i> -tagged jets, 99% $t\bar{t}$

Jet–boson categories. In *ee-Z* the hard and PU fractions of a jet depend strongly on the dilepton transverse momentum and on the azimuthal angle between the jet and the dilepton system: a jet recoiling against a boosted *Z* is hard in 96–99% of the cases, a jet in a $p_T^{\ell\ell} < 30$ GeV event is PU in 25–35% of the cases at low p_T and in up to 90% of the cases in the forward region. The jet table of *ee-Z* is therefore filled per jet in five categories (Table 2). Rows with different hard/PU mixtures are what separates w_H from w_{PU} in a given (η, p_T) cell; without the categories the two weights are degenerate ($\rho \simeq -1$).

Table 2: Jet–boson categories of the *ee-Z* jet table ($\Delta\phi$ between the jet and the dilepton system).

category	definition	PU fraction of DY jets, $25 < p_T < 50$ GeV	
		$ \eta < 2.5$	$ \eta > 3.0$
0	$p_T^{\ell\ell} < 30$ GeV	0.3–0.55	0.8–0.9
1	$p_T^{\ell\ell} \geq 30$ GeV, $ \Delta\phi < 1.5$		
2	$p_T^{\ell\ell} \geq 30$ GeV, $1.5 \leq \Delta\phi < 2.5$	↓	↓
3	$p_T^{\ell\ell} \geq 30$ GeV, $2.5 \leq \Delta\phi < 2.9$		
4	$p_T^{\ell\ell} \geq 30$ GeV, $ \Delta\phi \geq 2.9$ (balanced)	0.01–0.04	0.3

Fitted observables. Four histograms, all filled with raw data counts n_b :

1. the *ee-Z jet table*: number of jets per (category, η , p_T) cell, $5 \times 36 \times 8 = 1440$ bins;
2. the *ee-Z event table*: number of events per (N_{jet} class, $p_T^{\ell\ell}$ bin), $7 \times 26 = 182$ bins;
3. the μ -*W jet table*: jets per (η, p_T) cell, 288 bins;
4. the μ -*W event table*: events per N_{jet} class, 7 bins.

The jet grid uses the signed jet η with 36 bins (0.5 wide for $|\eta| < 2$, 0.25 for 2.0–2.5, 0.1 for 2.5–3.5, then 3.5–4.0 and 4.0–4.7) and 8 p_T bins with edges 25, 30, 35, 40, 50, 70, 100, 150 GeV, ∞ . The N_{jet} classes are 0, 1, $\dots$, 5, ≥ 6 ; the $p_T^{\ell\ell}$ bins have the edges

$$0, 2, 4, 6, 8, 10, 12.5, 15, 17.5, 20, 25, 30, 35, 40, 45, 50, \\ 60, 70, 80, 90, 100, 120, 140, 160, 180, 200 \text{ GeV},$$

with $p_T^{\ell\ell}$ capped at 200 GeV.

4 Model

Every simulated event e of family F in region R receives the weight

$$w_e = w_e^{\text{MC}} r(N_{\text{PV}}^{(e)}) \prod_{j \in e} w_{c(j)}(\eta_j, p_T^j) \times \begin{cases} k_{\text{DY}} g_{\text{DY}}(N_{\text{jet}}, p_T^{\ell\ell}) & F = \text{DY}, R = \text{ee-}Z \\ k_{\text{W}} g_{\text{W}}(N_{\text{jet}}) & F = \text{W}, R = \mu\text{-}W \\ \bar{g}_{\text{DY}}(N_{\text{jet}}) & F = \text{DY}, R = \mu\text{-}W \\ 1 & \text{otherwise,} \end{cases} \quad (1)$$

with the following ingredients.

Pileup profile, $r(N_{\text{PV}})$. The number of reconstructed primary vertices in simulation is reweighted to data, as a shape, in the *zero-jet* bin of ee- Z (bins of two vertices, clipped to $[0.05, 20]$), before anything else. The zero-jet bin does not depend on the jet corrections, and the reweighting removes the mismatch of the pileup profile that would otherwise be absorbed by the PU-jet weights. It is part of the correction and is deployed together with the maps; the same table is used for all regions and families.

Per-jet weights, w_{H} and w_{PU} . Two tables on the 36×8 grid, one applied to hard jets and one to PU jets ($c(j) \in \{\text{H}, \text{PU}\}$ from the generator matching). The event weight is the product over the jets of the event, so the expected content of a jet-table bin is a sum over jets of the event weight with the jet’s own factor divided out (“other-jet products”); the model is non-linear in the weights through these products. In $e\mu$ -Top the b -tagged jets are spectators (weight 1) and do not enter the jet observables. Cells without simulated jets of a given class (the 36 horn-veto cells and a few forward high- p_T cells) have their weight fixed to 1.

Process normalisations, k_{DY} and k_{W} . Free normalisations of the main family of each fit region; in the final tables they are quoted as the data/MC ratio of the zero-jet bin after all corrections, and in the analysis they are re-derived in every region from its own zero-jet bin, so that only the jet-dependent part of the corrections is transported.

DY theory correction, $g_{\text{DY}}(N_{\text{jet}}, p_T^{\ell\ell})$. A free table of 7×26 factors applied to the DY events of ee- Z (and, in the analysis, to the DY events of every dilepton Z region and the Higgs sidebands). Its class-0 row is the $p_T^{\ell\ell}$ shape of zero-jet events (its DY-yield-weighted mean defines k_{DY} and is divided out); the other rows carry the N_{jet} rates and their $p_T^{\ell\ell}$ dependence. It represents the mismodelling of the DY $p_T^{\ell\ell}$ spectrum and jet multiplicity (higher-order QCD, parton shower), which is a property of the process and not of the detector, and is therefore kept separate from the per-jet weights. Its $p_T^{\ell\ell}$ -marginal $\bar{g}_{\text{DY}}(N_{\text{jet}}) = \sum_b g_{\text{DY}}(n, b) m_{nb} / \sum_b m_{nb}$ (weights $m_{nb} = \text{DY yield without } g$) is applied to the DY contamination of the regions without a Z ($\mu\text{-}W$, $e\mu\text{-}Top$).

W correction, $g_{\text{W}}(N_{\text{jet}})$. Seven free factors (class 0 $\equiv k_{\text{W}}$) for the W+jets family in $\mu\text{-}W$, applied in every region to the W family.

Gauge. A common factor c on every jet weight is degenerate with $g(N_{\text{jet}}) \rightarrow g(N_{\text{jet}})c^{N_{\text{jet}}}$ in both fit regions. The convention of the previous procedure is kept: the hard-template-weighted mean of w_H over $|\eta| < 2.5$, $p_T \geq 50$ GeV (jets that are hard in $> 96\%$ of the cases) is 1, so that the “DY jet rate” is carried by g_{DY} and the per-jet weights describe only shapes and the PU rate. The gauge is imposed in the fit by a tight constraint on the weighted mean of $\log w_H$, and the exact linear convention is restored at export ($c = 1.0008$).

All parameters are fitted on the logarithmic scale, $\theta = \{\log w_H, \log w_{\text{PU}}, \log(k_{\text{DY}}g_{\text{DY}}), \log(k_Wg_W)\}$ (288+288+182+7 = 765 free parameters, plus the 7 derived $\log \bar{g}_{\text{DY}}$ factors of the W region, held at their marginal values and refreshed between rounds).

5 Likelihood

Grouped templates. With x_e the integer feature row of an event (hard-jet counts per cell, PU-jet counts per cell, its $(N_{\text{jet}}, p_T^{\ell\ell})$ or N_{jet} indicator), the weight of Eq. (1) is $w_e = w_e^{\text{base}} \exp(x_e \cdot \theta)$ with $w^{\text{base}} = w^{\text{MC}} r(N_{\text{PV}})$: the model is a log-linear Poisson regression. Events with identical x_e are grouped (two independent 64-bit hashes of the feature multiset, with an exact comparison of the feature rows inside every group), so that the expected content of bin b is

$$\mu_b(\theta) = \frac{1}{p_R} \sum_q A_{bq} e^{x_q \cdot \theta}, \quad A_{bq} = \sum_{e \in q} w_e^{\text{base}} h_{eb}, \quad (2)$$

where h_{eb} is the number of objects of event e in bin b (jets in a jet-table bin, 0/1 in an event-table bin), p_R the data prescale of the region, and A is a sparse matrix built in one streaming pass over the inputs (3.9 M groups from 70 M simulated events in ee- Z , 6.0 M groups from 126 M events in μ - W ; 7 minutes). The Jacobian $\partial\mu/\partial\theta = A \text{diag}(e^{x \cdot \theta}) X / p_R$ is exact; it was verified against finite differences to 10^{-7} . The same pass accumulates $A_{bq}^{(2)} = \sum_{e \in q} (w_e^{\text{base}})^2 h_{eb}^2$, which gives the statistical variance of the simulated prediction, $\sigma_{\text{MC},b}^2 = \sum_q A_{bq}^{(2)} e^{2x_q \cdot \theta} / p_R^2$.

Objective. The negative log-likelihood summed over the four tables,

$$-\ln L(\theta) = \sum_b \min_{\beta_b} \left[\mu_b \beta_b - n_b \ln(\mu_b \beta_b) + \frac{(\beta_b - 1)^2}{2s_b^2} \right] + \sum_i \frac{\theta_i^2}{2\sigma_i^2} + \frac{(\omega \cdot \theta_H)^2}{2\epsilon^2}, \quad s_b^2 = \frac{\sigma_{\text{MC},b}^2}{\mu_b^2}, \quad (3)$$

is the Poisson likelihood of the raw counts with the Barlow–Beeston-lite treatment of the simulation statistics (one Gaussian-constrained scale β_b per bin, profiled analytically: $\beta_b = \frac{1}{2}[(1 - \mu_b s_b^2) + \sqrt{(1 - \mu_b s_b^2)^2 + 4n_b s_b^2}]$; the relative variance s_b^2 is evaluated at the current template and refreshed between rounds, as the nominal-template convention of COMBINE’s `autoMCStats`). The priors are deliberately weak: $\sigma = 0.3$ on $\log w_H$, 1.5 on $\log w_{\text{PU}}$, 3 on the $\log g$ parameters, so that they act only in cells with (almost) no data; the last term is the gauge constraint (ω = normalised hard-template weights of the anchor cells, $\epsilon = 10^{-4}$). No region weights, no clipping of the weights and no exact constraints enter. Bins where the model is non-positive (negative generator weights in nearly empty cells) are clamped and removed from the gradient. The simulation statistics is essential: the fit without it does not converge and its goodness of fit on the ee- Z jet table is six times worse, because the hard-jet template statistics dominates the information of many cells.

Minimisation. Trust-region Newton (`scipy trust-exact`) with the analytic gradient and the Gauss–Newton Hessian $J^T \text{diag}(n_b / [\mu_b^2(1 + s_b^2 \mu_b)]) J$. From a cold start (all weights 1) the fit converges in 14 iterations with the original N_{jet} classes ($0, \dots, \geq 4$: 2 minutes per round, three rounds of the marginal/gauge refresh); with the classes extended to ≥ 6 the sparsely populated high- $N_{\text{jet}} \times p_T^{\ell\ell}$ cells slow the trust region down (300 iterations, 36 minutes per round) without moving any other parameter. A warm start from the previous procedure’s solution reaches the same physical solution (identical k , g and maps; the likelihood differs only in cells without data).

Goodness of fit. Saturated-model likelihood ratios, $2\Delta \ln L$: ee- Z jet table 2228 for 1259 populated bins, ee- Z event table 116/175, μ - W jet table 247/252, μ - W event table 2/7. The ee- Z jet-table excess is the residual 0.1-wide η structure inside the 0.5-wide central cells of the map (the same in the previous procedure, Pearson χ^2 2117 vs 2224 for the 252 cells summed over categories).

6 Uncertainties

Covariance of the parameters. The inverse of the Hessian of Eq. (3) at the minimum (with the penalty terms) gives the covariance Σ of all parameters: cells, $g_{\text{DY}}(N_{\text{jet}}, p_{\text{T}}^{\ell\ell})$, $g_{\text{W}}(N_{\text{jet}})$, k_{DY} , k_{W} , including every correlation. Fixed parameters (cells without simulation) have zero rows. Typical sizes: $\sigma(w_{\text{H}})/w_{\text{H}} = 0.8\%$ in the central region and 2–3% for $|\eta| > 3$; $\sigma(w_{\text{PU}})/w_{\text{PU}} = 7\%$ for central jets at 25–30 GeV, 1.3–3% in the forward region, $> 20\%$ above 100 GeV; $\sigma(\log g_{\text{DY}})$ 0.13/0.26/0.46/0.80% for the 1–4 jet marginals; $\sigma(\log g_{\text{W}})$ the same; the hard/PU weights of a cell are anticorrelated ($\rho = -0.67$ jet-weighted, -0.9 for central low- p_{T} jets); the correlation between $g_{\text{DY}}(\geq 3 \text{ jets})$ and the central low- p_{T} PU weight is only -0.25 (and -0.6 with the central PU weights above 50 GeV).

Sandwich estimator. The jet tables and the event tables of a region are filled by the same events, so the four histograms are not independent observations and the inverse Hessian is, in principle, optimistic. A second streaming pass at the fitted point accumulates the compound-Poisson covariance of the histogram vector of each region, $V = \sum_{\text{data}} h_e h_e^{\text{T}} + \sum_{\text{MC}} (w_e/p_R)^2 h_e h_e^{\text{T}}$, including the jet $\times$ event cross terms, and the sandwich covariance $\Sigma_s = H^{-1} G H^{-1}$ with $G = \sum_R J^{\text{T}} W V W J$ and $W = \text{diag}(1/[\mu_b(1 + s_b^2 \mu_b)])$, which reduces to H^{-1} exactly when V is diagonal. The result, σ_s/σ_H with median 0.996, 10th–90th percentiles 0.97–1.01 and a maximum of 1.11, shows that the overlap is negligible. Σ_s is the covariance exported with the maps.

Propagation to histograms. For any histogram h of a corrected simulation the band is the first-order propagation $\sigma_h^2 = \text{diag}(J_h \Sigma_s J_h^{\text{T}})$ with $J_h = \partial h / \partial \theta$: for a cell parameter, the histogram of the (jet $\times$ plotted object) pairs weighted by the event weight (the per-jet product makes $\partial h / \partial \log w_m$ equal to the contribution of the events with a jet in cell m , counted once per such jet); for a $\log g_{\text{DY}}(n, b)$ or $\log g_{\text{W}}(n)$ parameter, the histogram of the DY (W) events of that class; for regions that use the $p_{\text{T}}^{\ell\ell}$ -marginal, the chain rule $\partial \log \bar{g}_n / \partial \log g_{nb} = g_{nb} m_{nb} / \sum_b g_{nb} m_{nb}$ (minus the class-0 row that defines k_{DY}). When the ratio pad is shape-only (simulation scaled to the data yield) the Jacobian is projected onto shape variations. Figure 2 shows that the joint propagation matters in both directions: the η bands shrink by about a factor 2 with respect to a per-cell diagonal treatment (which added the anticorrelated σ_{H} and σ_{PU} of a cell in quadrature), while the N_{jet} and $p_{\text{T}}^{\ell\ell}$ bands grow because the covariance of g is now included. For the analysis, Σ_s is diagonalised and its leading eigenvectors (coherent variations of maps and g together) become correlated nuisance parameters, as for a reduced set of jet energy scale sources.

Non-closure. The two validation regions expose what the model cannot describe: the Top light jets are 2–2.5 times more often hard than Z/W jets at the same (η, p_{T}) , and $\mu\mu$ - Z has no free parameter. After the corrections, the shape-only data/MC ratio of the non- b jets on a coarser $(|\eta|, p_{\text{T}})$ grid is $1 + \delta_b$ per region, and every jet of a region’s family receives $1 \pm \delta_b$ as one fully correlated nuisance per validation region (Top: jet-weighted mean $|\delta| = 0.034$, $\mu\mu$ - Z : 0.015; 11 cells with $|\delta| > 0.2$, all at the $|\eta| = 3$ boundary or above 100 GeV in the forward region). These are shape nuisances; the normalisation of every region is fixed by its own zero-jet bin.

7 Results

Per-jet weights. Figure 1 shows the maps. The PU weights are far below 1 for central soft jets (0.51 at 25–30 GeV, 0.75 at 30–35, 1.0 at 35–40, rising to 1.1–1.15 above 50 GeV: the pileup jets at low p_{T} are over-predicted by a factor two) and 0.4–0.6 in the forward region at low p_{T} ; the hard

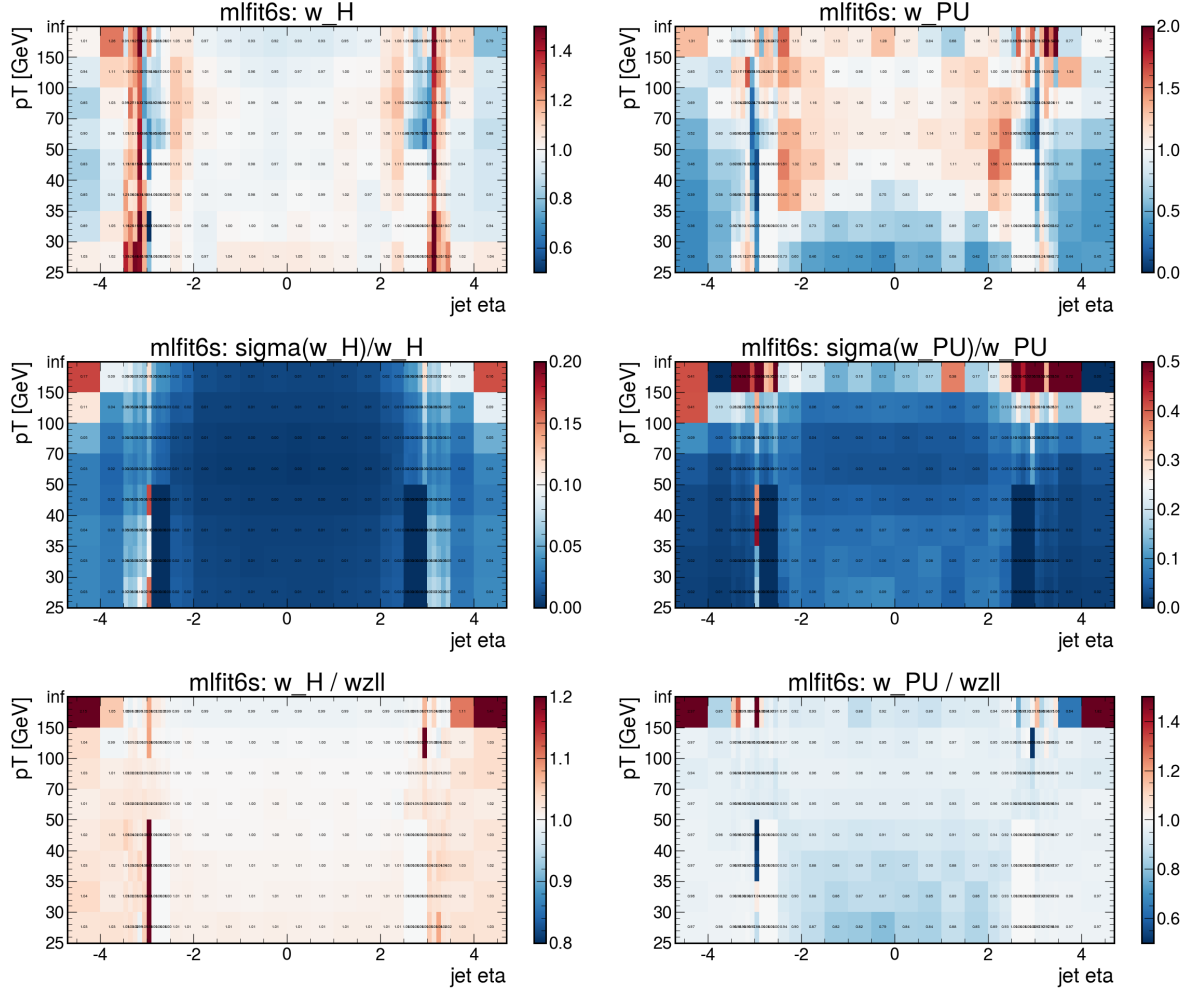


Figure 1: Fitted per-jet weights on the (η, p_T) grid: w_H (top left), w_{PU} (top right), their relative uncertainties (middle row) and their ratios to the weights of the previous (weighted least-squares) procedure (bottom row). The empty cells at $2.5 \leq |\eta| < 3.0$, $p_T < 50$ GeV are the horn veto.

weights are within 1–4% of unity in the central region and show the forward structure common to all processes: +5% at the HF edge (3.0–3.5) at the lowest p_T , –7 to –15% for $|\eta| > 3.5$ at high p_T . With respect to the previous procedure the PU weights are lower by 4–14% everywhere (most at central low p_T) and the hard weights higher by 1% (central) to 3% (forward); the previous g_{DY} values are 7–8 standard deviations away from the fitted ones, i.e. at the 2024 statistics the categories do separate the central PU rate from the N_{jet} correction, and the likelihood prefers less PU-jet suppression and a higher g_{DY} at ≥ 3 jets.

Process corrections. $k_{DY} = 1.0523$, $k_W = 1.0579$; the $p_T^{\ell\ell}$ -marginal and the W factors for $N_{\text{jet}} = 1, \dots, 5, \geq 6$ are

$$\begin{aligned}\bar{g}_{DY} &= 0.929, 0.895, 0.899, 0.992, 1.105, 1.286, \\ g_W &= 0.966, 0.959, 0.983, 1.114, 1.253, 1.417\end{aligned}$$

(previous procedure, classes to ≥ 4 : 0.928, 0.880, 0.869, 0.959 and 0.961, 0.954, 0.974, 1.133). The $g_{DY}(N_{\text{jet}}, p_T^{\ell\ell})$ table falls to 0.75–0.8 at $p_T^{\ell\ell} < 6$ GeV for 2–4 jets and is 0.85–1.0 above 30 GeV; the class-0 row (the zero-jet $p_T^{\ell\ell}$ shape) ranges from 0.9 at 4–6 GeV to 1.2 at 60–100 GeV.

Table 3: Shape-only χ^2 (data over the corrected simulation scaled to the data yield; data and simulation statistics) summed over the nine jet observables (N_{jet} , p_T and η of the two leading jets, m_{jj} , $|\Delta\eta_{jj}|$, all-jet η and p_T) per region, for the previous procedure (wzll) and this fit, without and with the band of Sec. 6 (fit covariance $\oplus$ non-closure nuisances). VBF = dijet subset.

region	stat wzll	stat this fit	ratio	+band wzll	+band this fit	ratio
μ -W	14803	14056	0.95	1030	957	0.93
μ -W VBF	1510	1531	1.01	658	632	0.96
$e\mu$ -Top	3126	3454	1.10	679	625	0.92
$\mu\mu$ -Z	18222	17720	0.97	979	859	0.88
$\mu\mu$ -Z VBF	1753	1727	0.99	556	495	0.89
ee-Z	14276	13877	0.97	1044	950	0.91
ee-Z VBF	1383	1339	0.97	539	480	0.89

Table 4: Selected observables: shape-only χ^2/n_{bins} without any band, with the fit covariance band, with the fit covariance $\oplus$ non-closure band, and the fraction of bins within one standard deviation of the full band (coverage).

region	observable	stat	+fit cov.	+non-closure	coverage
ee-Z	N_{jet}	59/8	32/8	5/8	0.75
ee-Z	$p_T^{\ell\ell}$	37/40	26/40	11/40	0.93
ee-Z	all-jet η	6083/94	5486/94	230/94	0.64
ee-Z	all-jet p_T	194/13	162/13	19/13	0.85
$\mu\mu$ -Z	N_{jet}	100/8	50/8	9/8	0.62
$\mu\mu$ -Z	$p_T^{\ell\ell}$	1007/40	485/40	246/40	0.53
$\mu\mu$ -Z	all-jet η	7257/94	6285/94	212/94	0.66
$\mu\mu$ -Z VBF	m_{jj}	118/40	95/40	68/40	0.57
$\mu\mu$ -Z VBF	$ \Delta\eta_{jj} $	112/40	92/40	50/40	0.57
μ -W	N_{jet}	4/8	2/8	1/8	1.00
μ -W	all-jet η	6369/94	5743/94	240/94	0.60
μ -W	all-jet p_T	124/13	108/13	21/13	0.85
$e\mu$ -Top	N_{jet}	228/8	214/8	96/8	0.12
$e\mu$ -Top	all-jet η	1187/94	1047/94	134/94	0.65
$e\mu$ -Top	all-jet p_T	394/13	315/13	12/13	0.77

Closure. Tables 3 and 4 and Figs. 2–4 summarise the validation. The corrected simulation describes the jet η and p_T spectra, the jet multiplicity and the dijet variables of the $Z \rightarrow \mu\mu$ region (not used in the fit) to the level of the band; the uncorrected simulation is off by factors (all-jet η χ^2 of 6.6×10^4 in $\mu\mu$ - Z). The N_{jet} distributions close in all Z and W regions with the N_{jet} classes extended to ≥ 6 (the $+6/ +20/ +26\%$ residuals at 5–7 jets of the earlier classes to ≥ 4 are gone). The W region closes as well as the Z regions, i.e. the per-jet weights are compatible between the two processes once each has its own N_{jet} correction. The $t\bar{t}$ region is where the model is incomplete: the light-jet p_T and N_{jet} (Table 4) are not covered by the fit covariance, because g is known to $< 1\%$ and Top has no process correction; the residual (a $\sim 5\%$ slope in N_{jet} and a p_T -dependent excess of forward jets) is the different hard/PU composition of the Top light jets and is carried by the Top non-closure nuisance. The overall picture is the same as with the previous procedure (it reproduces the earlier χ^2 to 3–5% in the Z/W regions and is 10% worse in Top), with the difference that every band is now derived from one likelihood.

8 Deployment and limitations

Deployment. Every simulated event of the analysis receives

$$r(N_{\text{PV}}) \prod_j w_{c(j)}(\eta_j, p_T^j) \times \begin{cases} g_{\text{DY}}(N_{\text{jet}}, p_T^{\ell\ell}) & \text{DY, } Z \text{ window and Higgs sidebands} \\ \bar{g}_{\text{DY}}(N_{\text{jet}}) & \text{DY elsewhere} \\ g_{\text{W}}(N_{\text{jet}}) & \text{W} \\ 1 & \text{otherwise,} \end{cases}$$

with the process normalisations re-derived per region from the zero-jet bin. The signal receives the per-jet weights (its tag jets are hard jets with weights within a few per cent of unity). The uncertainty enters as (i) the leading eigen-variations of Σ_s , each a coherent variation of maps and g , and (ii) the non-closure shape nuisances of the two validation regions.

Limitations and next steps.

- The hard/PU label is generator-level, so the weights are only defined for simulation; the anticorrelation $\rho(w_H, w_{\text{PU}}) \simeq -0.9$ for central soft jets is intrinsic and is carried by the covariance (the total jet yield of a cell is known to $\sim 1\%$, its split to 5–7%).
- The $t\bar{t}$ region is not described within the fit covariance; its residual is a model inadequacy (different hard fraction of the Top light jets), covered by a nuisance. Including the Top region in the likelihood with only its normalisation free is the natural extension.
- The W and Z regions disagree on the forward hard jets by 5–10% with opposite signs (a process-dependent effect, not a MET/selection effect); with its natural statistical weight the W region pulls less than in the previous procedure, and a W -specific forward hard-shape nuisance would make the maps independent of it.
- The per-cell priors could be replaced by a smoothness prior; with the present statistics they act only in empty cells.
- The maps are averaged over the 2024 eras; the later eras (higher pileup) show 10–15% larger PU deficits than the earlier ones.

A Software and reproducibility

Skim: `runners/multiflavor/` (`multiflavor_v4`, 312 condor jobs, merged on EOS); per-jet dumps: `dump_jets.py` (full statistics, W region with the corrected-MET selection and prescale 20). Fit:

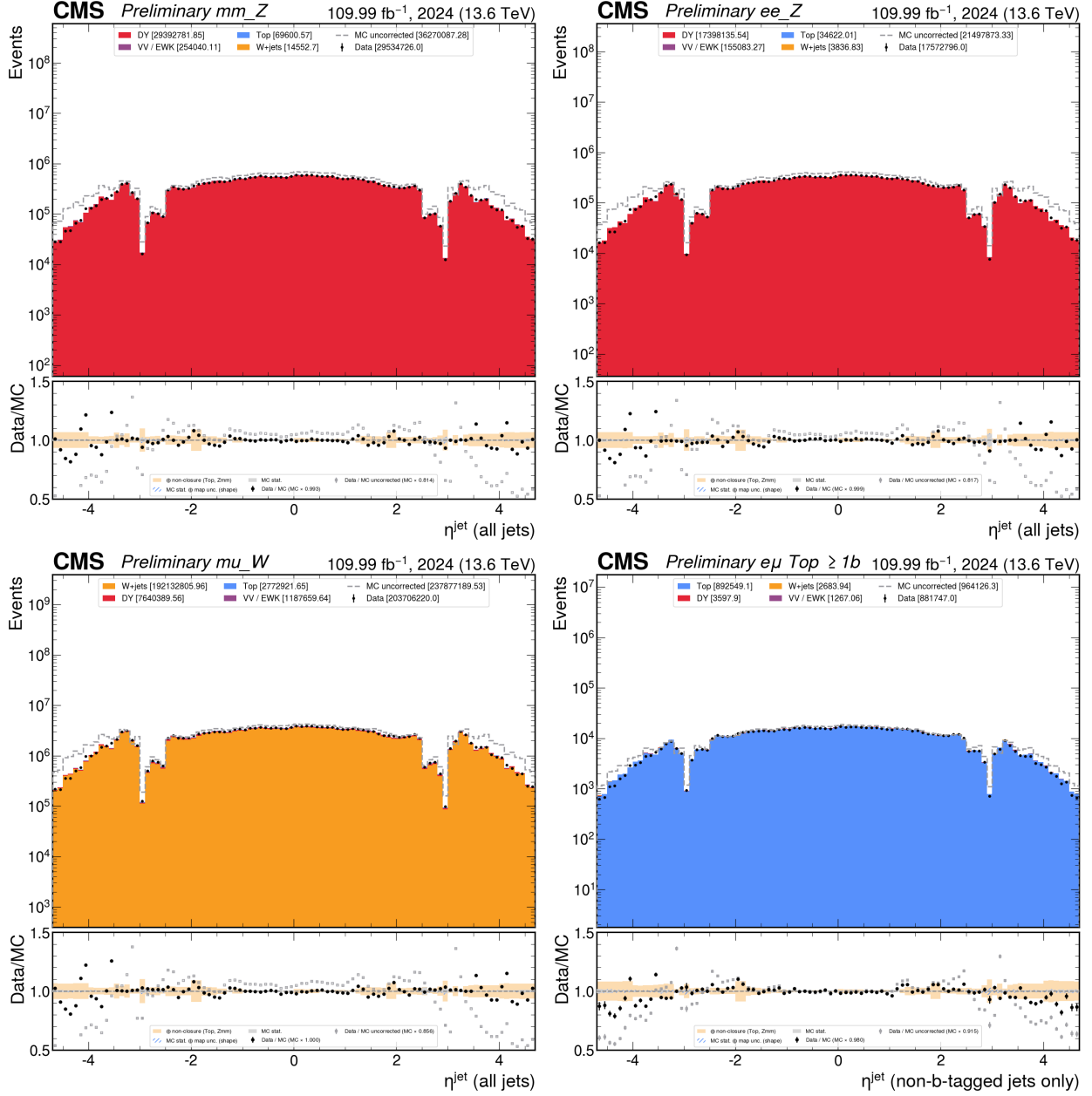


Figure 2: All-jet η in $\mu\mu$ -Z (validation), ee -Z (fit), μ -W (fit) and $e\mu$ -Top (validation, non- b jets). Points: data over the corrected simulation (black) and over the uncorrected simulation (grey squares, scaled to the data yield); bands: simulation statistics (grey), $\oplus$ the full fit covariance (hatched), $\oplus$ the non-closure nuisances (orange).

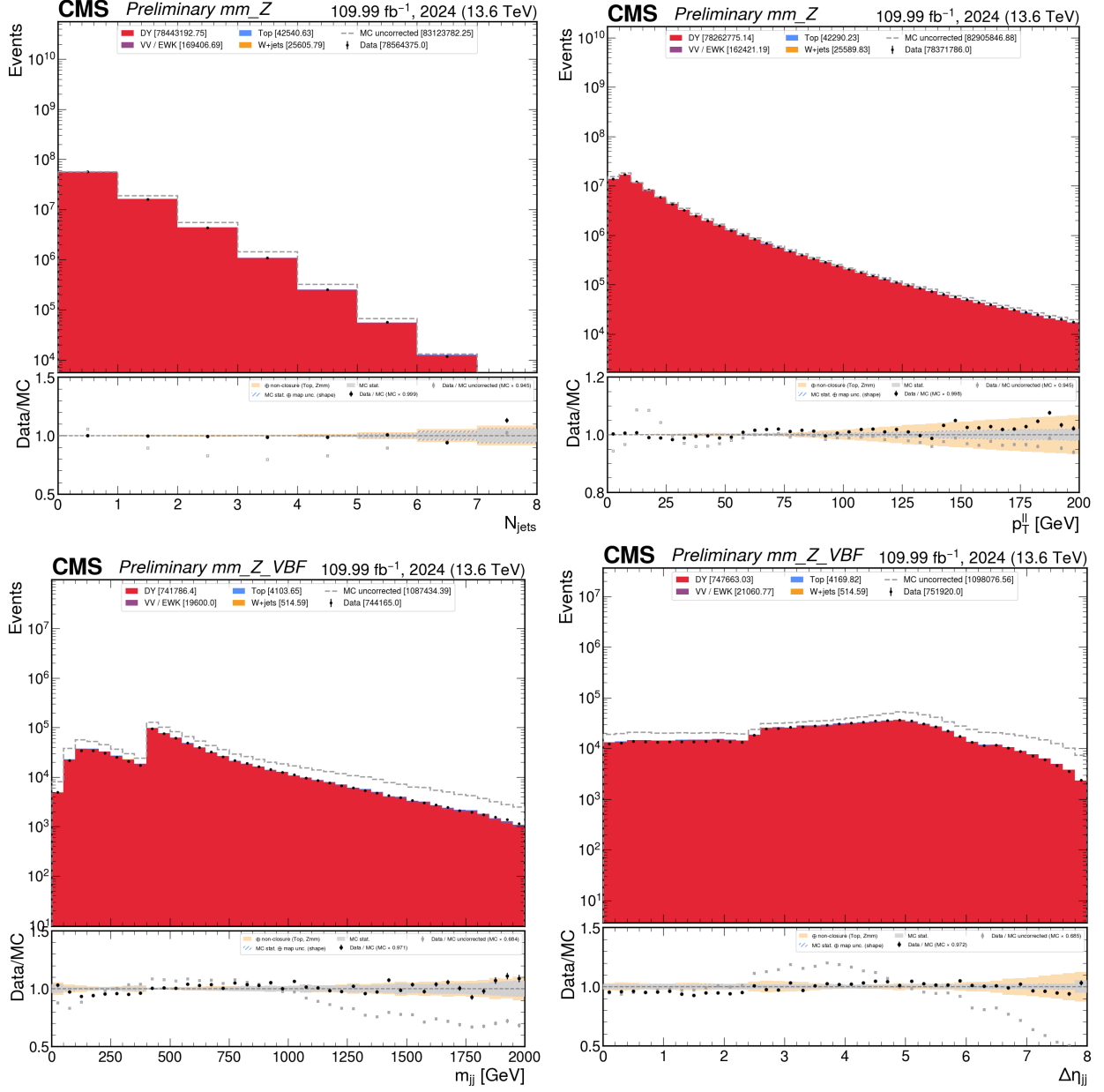


Figure 3: $Z \rightarrow \mu\mu$ (not used in the fit): N_{jet} , $p_T^{\ell\ell}$, and m_{jj} and $|\Delta\eta_{jj}|$ of the VBF dijet subset.

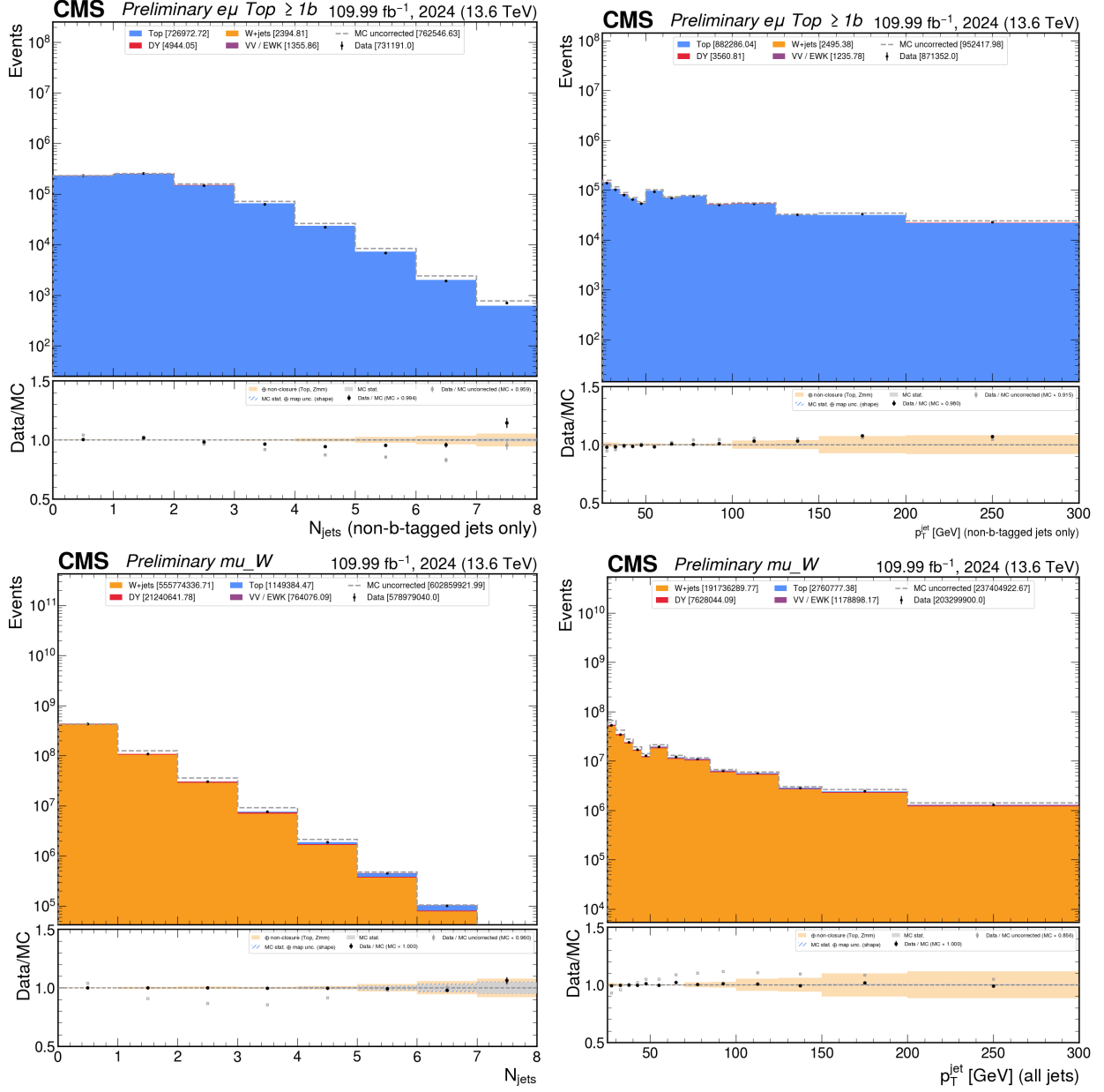


Figure 4: N_{jet} and all-jet p_T in the $t\bar{t}$ region (top, non- b jets; no process correction, the residual is covered by its non-closure nuisance) and in the W region (bottom).

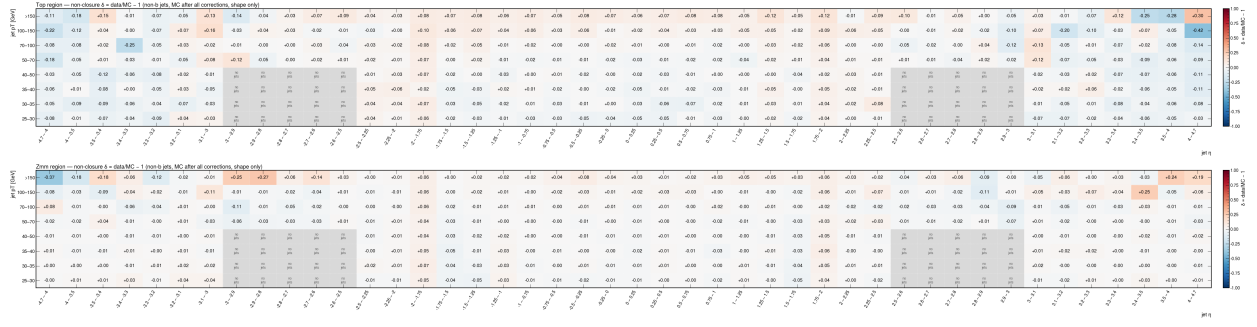


Figure 5: Non-closure δ_b of the non- b jets after the corrections, on the coarse $(|\eta|, p_T)$ grid, in $e\mu$ -Top and $\mu\mu$ -Z: the per-jet shape nuisances of Sec. 6.

runners/multiflavor/mlfit/ — build.py (templates and the covariance pass), fit.py (likelihood, minimisation, sandwich, export in the format of the previous chain), chain2.sh (the full sequence), compare_closure.py. Validation: the unchanged perjet_nonclosure.py and perjet_final_plots.py (with the --cov option for the full-covariance band). Results: analysis/plots_multiflavor_v4_perjet/mlfit6s/ (maps, g tables, covariance, summaries), web pages www/Hmm/multiflavor_v4_perjet/mlfit6s/; the chronological record is Sec. 5.25–5.26 of runners/multiflavor/PERJET_MAPS_REPORT.md. Run times on a 20-core machine: templates 7 min (10 GB cap), fit 2 min per round (36 min with the ≥ 6 classes), covariance pass 4 min, validation pages 1 h.